Air Pollution and Adolescents' Subjective Well-being: Evidence from South Korea

Abstract

We investigate the effect of exposure to particulate matter (PM_{10}) on subjective well-being of middle and high school students in South Korea. We control for individual fixed effects to rule out issues with residential sorting and unobserved time-invariant differences across individuals. Our findings suggest that the level of PM_{10} negatively affects the self-reported subjective well-being level of adolescents, with the effect being more concentrated among students with low-income households and male students. We also find a non-linear effect, indicating that the negative effect of particulate matter becomes weaker with a higher level of concentration.

Keywords: Air pollution, particulate matter, subjective well-being, adolescent

1 Introduction

Air pollution is a complex mixture of physical pollutants that affects human health through multiple pathways and is considered as one of the biggest health threats of our time (World Health Organization, 2016). While air pollution consists of various pollutants, including ozone (O_3), particulate matter (PM), carbon monoxide (CO), nitrogen oxides (NO_2), and sulfur dioxide (SO_2), studies have generally shown that PM affects human health to a greater degree than any other single pollutant (Hamanaka and Mutlu, 2018; World Health Organization, 2016). Additionally, PM is also known to be the main cause of visibility degradation by creating haze in the air. As human psychology and physiology are sensitive to visual input (Velarde et al., 2007; Mace et al., 2004), PM may bring about negative emotions such as stress and annoyance that may lead to lower levels of well-being.

In this study, we investigate the effect of exposure to particulate matter (PM_{10}) on subjective well-being (SWB) of middle and high school students in South Korea.[1] Two reasons make South Korea an adequate place to study the relationship of air pollution and SWB of adolescents. First, South Korea records one of the highest levels of fine particulate matter concentration among OECD countries in 2023, as reported in the Sustainable Development Report (Sachs et al., 2025). Second, South Korea has the highest suicide rate among OECD countries (OECD, 2024) and has a considerably high proportion of adolescents (15-year-olds) reporting dissatisfaction with life (22%) compared with the OECD average of 18% in 2022 (OECD, 2023). High level of academic stress driven by competitive education system in Korea (Lee and Larson, 2000) may intensify the negative psychological effects of environmental stressors such as air pollution.

In studying the impact of air pollution on SWB, we examine multi-dimensional aspects of SWB including life satisfaction, the positive affect of happiness, and the negative affect of anxiety and depression. The comprehensive measure of SWB includes both a cognitive and affective self-evaluation of one's life and experiences (Diener, 1994). Compared to affective measures that focus on current emotions and feelings, life satisfaction reflects the cognitive component of SWB

[1] Although $PM_{2.5}$ is known to have greater adverse health impacts than PM_{10}, we focus our analysis on PM_{10} due to a lack of data of $PM_{2.5}$. This is consistent with many previous studies that focus on air pollution on PM_{10} (Levinson, 2012; Montanari et al., 2025).

that evaluates individuals' lives as a whole and is known to be a more stable component of SWB (Suldo et al., 2006).[2] Consistent with previous literature, we use information from "happiness" and "life satisfaction" related questions (Dolan et al., 2008; MacKerron and Mourato, 2009) along with negative affect of anxiety and depression to measure SWB.

Our research contributes to several streams of literature. First, our research adds to understanding the overall impact of air pollution on human health. It is well documented that exposure to PM increases tangible health risks such as respiratory and cardiovascular diseases, as well as hospital admissions (Neas et al., 1994; Brook et al., 2004; Pope III et al., 2009; EPA, 2009; Hicken et al., 2014), and that it is less well known how air pollution affects subjective well-being (SWB) and mental health. We add to this literature by investigating these intangible outcomes, providing a more comprehensive measure of well-being. Additionally, our research enriches the growing literature on positive psychology and happiness economics, which suggests that non-income factors are crucial for accurately evaluating the costs of air pollution (Gowdy, 2004; Welsch, 2009).

Second, our study adds to the literature on the effect of pollution on SWB. While several studies have demonstrated the detrimental effect of air pollution on SWB (Levinson, 2012; Orru et al., 2016; Li et al., 2024), there is a lack of study that focuses on adolescents. Adolescence is a critical stage in life where both internal and external changes take place: internally, vast hormonal and biological changes take place, and externally, dramatic shifts in social relationships with peers and parents take place (Guyer et al., 2016). As life course theory posits that life stages are interdependent, healthy adolescence may be pivotal in forming a healthy and happy life in later adulthood (Elder Jr et al., 2003). In fact, studies show that positive well-being in adolescents is related to a lower risk of illness and mortality in later life (Pressman and Cohen, 2005; Dockray and Steptoe, 2010).

Third, our study expands the literature on the determinants of SWB of adolescents, from a traditional emphasis on factors of academic achievement (Steinmayr et al., 2016), social support including peers and family (Siedlecki et al., 2014), and family structure (Yárnoz-Yaben and Gar-

[2]The term "happiness" is often used as a convenient shorthand for SWB, but happiness is only a part of SWB which is a more comprehensive concept that includes life satisfaction, positive affects (e.g. happiness, joy, excitement), and negative affects (e.g. pain, sadness, anxiety) (OECD, 2025).

mendia, 2016) to an environmental factor such as air pollution. Although a growing body of research has begun to examine the effects of air pollution on adolescent mental health (Ge and Zhang, 2025; Jiang et al., 2023), to our knowledge, empirical evidence remains limited, especially regarding happiness or subjective well-being. Lin et al. (2019) is a notable exception, using a Taiwan cohort study to document a negative effect of air pollution on adolescent happiness. However, this study relies on cross-district variations in pollution for identification, which can result in nonrandom assignment of pollution to adolescents under residential sorting (Graff Zivin and Neidell, 2013).

The common challenge of analyzing the relationship between air pollution and SWB is assigning high-quality data on air pollution with detailed spatial disaggregation to the location of individuals at the time of the survey. Rehdanz and Maddison (2008) use the self-reported adverse impact of air pollution, rather than using actual pollution data, to establish a negative correlation with SWB. We use detailed spatially disaggregated pollution data, which is an advantage compared to studies that rely on a country level aggregate pollution data (Welsch, 2006; Menz, 2011). Abed Al Ahad (2024) notes that there is lack of research on the mediating effect of general health on the association between long-term exposure to air pollution and LS using longitudinal individual-level data linked to fine spatial resolution air pollution data for multiple pollutants.

Even with the precise assignment of pollution levels to individuals, the mobility of individuals (not in response to pollution) may create a measurement issue. Individuals move over time to different locations to their work, home, and possibly to other cities, which makes it difficult to avoid mobility bias (Graff Zivin and Neidell, 2013). However, our study is less susceptible to this mobility bias, as Korean middle school and high school students mostly attend schools within their residential area due to the 'equalization policy' that has been implemented from 1974. This policy randomly allocates students to schools within their residential district to prevent overheated competition to enter prestigious high schools. The fact that the students are much less mobile since they mostly spend their time in their school located within their residential area increases the preciseness of assigned pollution levels. Furthermore, the longitudinal nature of the survey enables us to overcome estimation biases that may occur due to differences in individual

time-invariant characteristics, by including individual fixed effects.

Our findings suggest that the level of PM_{10} negatively affects the self-reported SWB level of adolescents, with the effect being more concentrated among students with low-income households and male students. We also find a strong non-linear effect, indicating that the negative effect of PM_{10} becomes weaker with a higher level of concentration.

2 Data

To investigate the impact of particulate matter on SWB of adolescents, we use the Korean Children and Youth Panel Survey in 2010-2016 (KCYPS 2010), air pollutant data on the *Airkorea* website,[3] and weather data by the Korea Meteorological Administration. The KCYPS 2010 was conducted by the National Youth Policy Institute, with the aim of identifying various aspects of the growth and development of children and youth and as such, it contains rich information including student's SWB and its determinants (National Youth Policy Institute, 2010). The survey design employs the stratified multi-stage cluster sampling method. A multiple point perspective panel design was used by repeatedly tracking the three cohorts in the first year of survey: 1st grade, 4th grade, and 7th grade (equivalent to US grade levels). Our study focuses on analyzing middle and high school students during the survey period, which includes students from 7th to 12th grade.[4]

We match KCYPS 2010 data with air pollution data and weather data by district and survey period. Ministry of Environment of Korea has been releasing data on air quality on *Airkorea* website since December 2005. *Airkorea* publishes real-time Comprehensive Air-quality Index (CAI) and concentrations of major air pollutants (such as PM_{10}, $PM_{2.5}$, O_3, CO, SO_2, NO_2) for all state-controlled monitoring stations. We compute the average air pollution data for each monitoring station during the survey period by taking the mean of the hourly values. Since the survey was taken during the months of October through December at an hourly basis, we use pollution data

[3] `www.airkorea.or.kr`

[4] Our analytical sample consists of students from the second and third cohorts, restricted to district-year combinations with more than 10 observations to ensure statistical reliability of the estimates. We also exclude observations from 2011 because SWB information is not available in that year.

from October through December as an approximation of the pollution experienced during the school year. In Figure 1, the distribution of average PM_{10} shows a normal distribution and exhibits a similar pattern across three months. To obtain the pollution measure for each student, we average the pollution levels for each residential district of the student.[5] Students living in districts that do not have monitoring stations were dropped from the study. Figure 2 shows the location of districts with air pollution monitoring stations.

[Figure 1]

We also include rich weather data to control for weather patterns that may affect the impact of air pollution. For weather variables, we use the Automatic Weather System (AWS) data managed by the National Weather Data Center.[6] The AWS has about 510 locations (as of January 2017) nationwide installed to automatically observe weather conditions, which observe temperature, precipitation, wind, humidity, and air pressure. We average hourly data on temperature, precipitation, and the average and maximum wind speed for each location during the survey period.

[Figure 2]

2.1 Subjective well-being variables

The measures of SWB in the KCYPS include overall life satisfaction, positive affect, and negative affect (Diener et al., 2003; OECD, 2025). The measurement was carried out using a self-reported four-point Likert scale: number one on the scale indicates 'strongly agree', two indicates 'somewhat agree', three indicates 'somewhat disagree', and four indicates 'strongly disagree'. Each SWB measure adopts this scale to the following statements: for cognitive evaluation of overall life satisfaction, "I think my life is happy"; for positive affect, "I enjoy living my life". Answers for life satisfaction and positive affect are reverse coded so that a higher score on overall life satisfaction and positive affect reflects a higher level of SWB.

[5] "District" is defined as the second-level administrative division including 262 local governments (Si/Gun/Gu) and "province" is defined as a group of 17 first-level administrative divisions including 9 provinces, 6 metropolitan cities, 1 special city, and 1 special autonomous city.

[6] Compared to Automated Synoptic Observing System (ASOS) for weather data, the AWS has more weather observation stations which allows us to match more districts for each individual.

For negative affect, we use questions related to anxiety and depression. The questions associated with each variable are detailed in Table A1.[7] Since responses to these questions are likely to be correlated, we use principal factor analysis to create an index that takes a continuous value between 0 and 1. Factor loadings used to create anxiety and depression variables are shown in Table A2. A higher score on the negative affect indicates a lower level of SWB.

2.2 Summary statistics

Table 1 describes key variables and their summary statistics. The sample size for all years is 12,374. Panel A shows the summary statistics for four outcome variables of SWB.

Panel B reports the time students spend on various daily activities: sleeping, private-tutoring, studying, reading, engaging in digital-entertainment, and playing with friends after school. The average time a student spends sleeping is about 8.2 hours a day, attending private-tutoring is about 1 hour, and studying (such as doing homework) is about 2.3 hours. Reading time is approximately 0.5 hour, digital-entertainment time is about 3.3 hours, and time spent playing with friends is about 1.5 hours.

Panel C shows the yearly income in the unit of 10 million KRW (about 10,000 USD)[8] and the average yearly income of the students' household is 46 million KRW. The average parents' years of education is 14.2.[9] Other control variables include the sum of peer relationships, the level of supportive parenting, and the frequency of abuse, each measured on a 4-point Likert scale. The extent of victimization is a variable constructed by summing up the responses to each victimization question, where a response of 1 indicates the presence of an experience, and a response of 0 indicates its absence.[10]

The summary statistics of the five pollution variables including PM_{10} is shown in Panel D.

[7] The anxiety and depression indices are available only for the selected waves, resulting in a smaller estimation sample (N = 8,116) compared to the life satisfaction and happiness outcomes (N = 12,374).

[8] 1,000 KRW is worth approximately 1 USD. The KRW–USD exchange rate ranged from 1,055.4 to 1,207.7 during the period 2010–2016 (Source: Ministry of Economy and Finance, `https://www.index.go.kr/unity/potal/main/EachDtlPageDetail.do?idx_cd=1068`).

[9] Years of education are assigned as follows: 6 years for an elementary school graduate, 9 years for a middle school graduate, 12 years for a high school graduate, 14 years for a 2-year college graduate, 16 years for a 4-year college graduate, 18 years for a master's degree graduate, and 23 years for a PhD degree graduate.

[10] For more details on these variables, refer to Table A1.

The average concentration of PM_{10} is approximately 47.0μg/m^3 throughout 2010–2016. This is close to the highest annual mean level of Interim Target-2 (IT-2) set by the WHO Air Quality Guidelines (Table A3). Panel E shows that the average temperature measured in degrees Celsius is 8.4, the average rainfall measured in millimeters is 1.9, and the wind speed measured in meters per second is 1.9.

[Table 1]

3 Empirical model

To examine the relationship of PM_{10} on SWB, we estimate the following equation:

$$y_{ijt} = \alpha + \beta_1 PM_{10\,jt} + \beta_2 PM^2{}_{10\,jt} + W'_{jt}\gamma + X'_{it}\delta + \sigma_i + \tau_t + \epsilon_{ijt} \tag{1}$$

The dependent variable y_{ijt} is the SWB of respondent i in district j at year t. The independent variable $PM_{10\,jt}$ is the adjusted particulate matter measure in district j at year t. Building on prior evidence of non-linear relationships between air pollution and subjective well-being (Tian et al., 2022), our primary specification includes both linear and quadratic terms of PM_{10} to flexibly capture the dose-response relationship with SWB. We divide the actual value of PM_{10} by 10 so that a unit of $PM_{10\,jt}$ is measured in 10 μg/m^3. For comparison purposes, we also report results from a linear specification (excluding the quadratic term) in our baseline analysis, but we focus our discussion on the quadratic specification throughout the paper.

The vector W'_{jt} includes the weather control variables (temperature, wind speed, rainfall) in district j at year t. The vector of X'_{it} includes the individual characteristics for respondent i at year t, which are selected based on previous literature on SWB of adolescents. First, we include peer relations as controls, as well as the level of victimization individuals experienced during school and social life. Positive social support from peers is known as an important factor to increase SWB by empowering adolescents to better cope with difficulties in life (Siedlecki et al., 2014; Bi et al., 2021). We also include the level of supportive parenting and abuse experienced

from their parents, as parenting style is another factor that may influence the SWB of adolescents (Garcia et al., 2019; Aquilino and Supple, 2001). Adolescents raised with higher parental warmth and lower parental strictness have been found to have greater SWB (Garcia et al., 2019). Finally, the demographic variables such as age, average yearly income, and parent's years of education are included as additional control variables.

Equation 1 further includes the individual fixed effect σ_i and the year fixed effects τ_t. ϵ_{ijt} is the idiosyncratic error term. Standard errors are clustered at the district level. The key identifying assumption to infer a causal relationship between particulate matter and SWB is that the unobserved determinants of an individual's SWB are uncorrelated with the pollution level. To indirectly test this assumption using observable characteristics, we conduct a balance test by replacing the dependent variable with each of the control variables in Equation 1.[11] The results are presented in Table A4, with and without individual fixed effects. Table A4 shows that while several variables show significant relationship with pollution when estimated without individual fixed effects (panel A), Panel B, which includes individual fixed effects, shows no statistically significant relationship. This test indicates that relying on cross-sectional variations in pollution may result in biased estimate and that, conditional on individual fixed effects, the remaining within-individual variation over time in pollution is more likely to be orthogonal at least to observable characteristics.

Several empirical issues exist in estimating the causal effect of pollution on individual well-being outcomes, including accurately assigning pollution to individuals discussed in the Introduction (Graff Zivin and Neidell, 2013). The first issue is residential sorting. For example, the nonrandom assignment of pollution levels can occur if wealthier individuals live in less polluted area due to their preferences for local amenities, and these individuals also make other investments to improve their SWB. Then the negative effect of PM on SWB would be exaggerated. On the other hand, individuals may live in more polluted areas because of commuting cost and job opportunities, and if they tend to enjoy more activities to improve SWB, the negative effect of air pollution would be understated. Our approach helps mitigate this nonrandom assignment problem of pollution by controlling for unobserved time-invariant individual fixed effects and

[11] As we later show in Table 3, the variables included for the balance check significantly predict SWB.

income (Graff Zivin and Neidell, 2013).[12]

Second, weather and other environmental factors (pollutants other than PM) may be confounders that can be correlated with the PM and SWB. We include weather variables in the main regression and report the robustness of our results when including pollutants other than PM, such as NOx and Ozone.

Finally, individuals may take deliberate actions to avoid pollution in their everyday lives by buying air purifiers, wearing face masks (i.e. market based) and choosing to stay indoors for most of the day after they find out the level of pollution on the weather forecast (i.e. non-market based). Since we do not directly observe this avoidance behavior, our results are likely to reflect the overall reduced-form effect of pollution; i.e. the effect after individuals optimize their behavior in response to pollution given their resources. This avoidance behavior would attenuate the effect of PM on SWB relative to a counterfactual scenario in which such behavior is not possible.

4 Results

4.1 Baseline results

Table 2 presents the baseline estimates of the relationship between PM_{10} and subjective well-being (SWB) based on Equation 1. Model (1) (Columns 1–3) presents the linear specification, including only the linear term of PM_{10}, while Model (2) (Columns 4–6) presents the non-linear specification, including both linear and quadratic terms of PM_{10}. Within each model, we progressively add control variables: Columns (1) and (4) include no control variables, Columns (2) and (5) add individual and year fixed effects, and Columns (3) and (6) includes the weather variables and all the individual control variables that were selected as determinants of adolescents' SWB. We focus our interpretation on Columns (6), which include the full set of controls.

The estimates of air pollution in Table 2 show the results on the relationship between PM_{10}

[12]In the robustness check, we construct the analysis sample using only students who did not relocate during the survey period, thereby further eliminating the possibility of residential sorting. The results remain similar to the main analysis.

and subjective well-being. Panel A shows results for life satisfaction. The coefficient on PM_{10} is estimated to be negative and significant in Column (1) (-0.025, p<0.01) but becomes insignificant with individual and year fixed effects (-0.020, p<0.01 in Column 1, insignificant in Columns 2–3). Panel B presents similar patterns for happiness (-0.020, p<0.01 in Column 1, insignificant in Columns 2–3). Recent studies document non-linear relationships in the pollution-subjective well-being or mental health decline nexus (Tian et al., 2022; Jung et al., 2019). If similar non-linearity is present in our data, the linear specification may average across positive and negative effects, possibly resulting in an insignificant coefficient. To test for potential non-linearities, we next examine a quadratic specification.

The estimates of air pollution in Model 2 of Table 2 show a negative and significant relationship on SWB. The increase in the value of PM_{10} by 10 decreases life satisfaction roughly by 0.093, while this negative effect decreases in magnitude as captured by the positive coefficient on the squared term of PM_{10}. Given that the average value of life satisfaction is 3.1, an increase in PM_{10} by 10 $\mu g/m^3$, say from the average PM_{10} of 47.0 to 57.0 translates into about 0.27 percent decrease in life satisfaction of the baseline outcome. This is consistent with the findings of Li et al. (2021), who document adverse effects of air pollution on noncognitive traits among Chinese adolescents. Using the Air Pollution Index (API) — a composite measure of multiple pollutants including PM_{10} — they find that a 15-point increase in API leads to a 3.2% reduction in self-satisfaction (measured as a noncognitive trait on a 0–4 scale) and a 5.5% increase in psychological distress. The study also confirms that air pollution has a significantly greater impact on psychological distress in adolescents compared to young adults, middle-aged, and elderly groups. The effect of air pollution on happiness is also negative, and statistically significant. This is in line with recent literature measuring the impact of particulate air pollution on hedonic happiness (Zhang et al., 2017; Lin et al., 2019).

[Table 2]

The positive coefficient for the squared term of PM_{10} indicates a nonlinear relationship between life satisfaction and pollution. This non-linear relationship is consistent with previous research that shows an inverted U-shaped relationship between high exposures to PM_{10} and

mental health deterioration (Jung et al., 2019). Figure A2 depicts the marginal effect of PM_{10} after taking into account this non-linear effect. The standard errors are calculated by the delta methods. The negative effect of PM_{10} on subjective well-being attenuates at different thresholds: around 55 $\mu g/m^3$ for life satisfaction and around 63 $\mu g/m^3$ for happiness, beyond which the effects are no longer statistically significant. This could be because at high levels of pollution, individuals' avoidance behaviors, such as responding to air pollution alerts, might mitigate the effects.[13] At the same time, the attenuation at higher pollution levels could also partly reflect reduced statistical precision due to fewer observations in the upper tail of the PM_{10} distribution (see Figure 1).

Table 3 reports regression results with control variables and further examines the effects of PM_{10} on negative affect outcomes, including anxiety and depression. The control variables include as other determinants of SWB are in line with previous literature. Specifically, consistent with SWB literature and empirical results measuring the effect of air pollution and happiness (Zhang et al., 2017; Daly, 2022), there is a negative relationship between age and life satisfaction where students or young and middle-aged adults that are older tend to report lower levels of life satisfaction. The relationship of life satisfaction and control variables are also consistent with previous literature suggesting that peer relation and supportive parenting contribute positively to SWB, while higher levels of perceived abuse and victimization contribute negatively to SWB of adolescents.

Next, we investigate the effect of PM_{10} on SWB measures of negative affect including anxiety and depression in Table 3. Increases in PM_{10} are associated with higher levels of anxiety. This finding is consistent with growing evidence on air pollution's adverse effects on adolescent mental health. Ge and Zhang (2025) find that each 1 $\mu g/m^3$ increase in $PM_{2.5}$ concentration was associated with a 0.319-point increase in the CES-D8 depression score among Chinese adolescents. Similarly, Zundel et al. (2022), in a systematic review of the MEDLINE database conducted in September 2021, find that air pollution exposure is consistently associated with adverse effects on internalizing psychopathology, including anxiety and depression. For depression (column 4), our coefficient is positive but imprecisely estimated and not statistically significant. This may

[13] Jung et al. (2019) find that Korean households limit outdoor activities when pollution levels are high.

reflect the fact that depression, as a more persistent condition, is less responsive to contemporaneous pollution fluctuations than acute stress or anxiety.

[Table 3]

A possible mechanism through which particulate matter affects SWB is that pollution alters students' time use in a way that correlates with their SWB. For instance, as part of individuals' optimizing behavior in response to pollution, students may spend more time indoors when pollution levels are high, altering their time use and potentially affecting SWB. To explore this possibility, Table 4 shows whether particulate matter affects the time use of students. Although we do not have a direct measure on whether the student stays indoors, the KCYPS data include information on times of sleeping, private tutoring, self-study, reading, digital-entertainment, and time spent with friends.[14]

[Table 4]

The results in Table 4 show that while PM_{10} reduces the average reading time of students, it does not significantly impact other types of activities. Evidence from the literature indicates that exposure to air pollution negatively affects students' cognitive functioning (Lopuszanska and Samardakiewicz, 2020; Forns et al., 2017). This impact may particularly lead to a reduction in activities that are not essential but contribute to academic development, such as discretionary reading time.

Additionally, although the coefficient is not statistically significant in the full sample, we observe a negative association with sleep duration. As will be detailed in Table 6 in the following section, we find statistically significant reductions in sleep time particularly among male students. Exposure to ambient air pollutants has detrimental effects on sleep quality and sleep

[14]Sleeping time refers to the average sleep hours per day. Private tutoring time is the time spent at private academies or with tutors and study time means how much students spend on homework or studying during the day. Reading time refers to the time spent reading books other than textbooks or reference books. Digital-entertaining time includes time spent on computers, playing video games, and watching TV or DVDs, and friend time means how much time the student spends playing with friends (excluding doing computer or playing games, or watching TV) after school during the day. We use the average value of each time during the weekdays and weekend.

duration (Cao et al., 2021; Chen et al., 2019; Liu et al., 2020) and it affects sleep-disordered breathing (wheezing and snoring) in children (Sánchez et al., 2019; Lawrence et al., 2018). Sleep quality and duration are among the factors that significantly impact life satisfaction (Ness and Saksvik-Lehouillier, 2018; Pagan, 2017), and these effects are likely to contribute to a reduction in overall life satisfaction.

4.2 Heterogeneous effects

Our main results show the overall influence of PM_{10} on SWB. In this section, we investigate whether the effects differ by gender and socio-economic status (SES) for the following reasons. First, male children generally report a higher level of life satisfaction than female children in Korea (Kim and Jeong, 2017), suggesting that the effect of particulate matter may differ by gender. Our data indeed shows a gender difference in the effect of PM_{10} and life satisfaction (Table 5). When PM_{10} increases, the life satisfaction of boys significantly decreases, but the life satisfaction of girls is not affected by PM_{10}. Given this gender difference, boys may respond differently to pollution than girls. Our findings are consistent with prior evidence suggesting that males may be more sensitive to air pollution than females. For instance, Ebenstein et al. (2016) document larger adverse effects of air pollution on boys' cognitive performance.

[Table 5]

Additionally, Table 6 shows the impact of particulate matter on students' time use by gender. It can be observed that air pollution has a significant effect on reducing sleeping and reading time for male students, while for female students, an increase in air pollution corresponds to an increase in private tutoring time. A 10 $\mu g/m^3$ increase in PM_{10} is associated with a reduction of approximately 30 minutes in daily sleep duration among male students. Yu et al. (2019) examine Chinese university students in Beijing and find that a one standard deviation increase in air pollution concentration is associated with a 0.5–0.7 hour reduction in daily sleep duration, which is comparable in magnitude to our estimates.

[Table 6]

As a second dimension of heterogeneity, previous literature suggests that vulnerable populations, including those with lower income, are more susceptible to pollution (Laurent et al., 2007; Ebenstein et al., 2016). To investigate this, we examine the heterogenous effect by SES in Table 7.

[Table 7]

We classify students into 'Low SES' and 'High SES' groups based on whether their household income falls below or above the median of the income distribution for each year. Our findings reveal that air pollution affects time allocation differently across SES categories, with distinct heterogeneous patterns emerging. In particular, for the 'Low SES' group, private tutoring and study time decrease by approximately 0.2 and 0.5 hours per day, respectively, both statistically significant. In contrast, the 'High SES' group experiences a 0.3-hour increase in private tutoring time. The result implies that children from low SES backgrounds may experience an even greater negative impact on study time with increased levels of pollution, given that these children already spend less time studying on average compared to those from high SES backgrounds.

[Table 8]

4.3 Robustness Checks

In this section, we present three robustness checks: (1) comparing subsamples defined by smaller versus larger geographic areas, (2) restricting the sample to students who did not relocate during the survey period, and (3) including additional pollutant variables. We conduct these robustness checks using life satisfaction and happiness as representative measures of SWB.

First, since smaller geographic areas are more likely to be located near monitoring stations than larger areas, the size of the area in which individuals reside in may introduce measurement error in air pollution exposure. To address this concern of measurement error, we report the results based on the size of the district to ensure a precise match. Specifically, we split the sample based on whether the observations are located within the first-level administrative divisions in South Korea,[15] which are densely populated but much smaller in area than other administrative

[15]These locations include 1 special city (Seoul), 6 metropolitan cities (Busan, Daegu, Incheon, Gwangju, Daejeon, Ulsan), and 1 special self-governing city (Sejong).

divisions. Columns (1) and (2) in Table 9 show the results. The results indicate that negative and significant effect of pollution holds true for both subsamples regardless of the size of area, but the subsample of the smaller sized area, which has a higher population density, has a higher negative effect in magnitude than the subsample with larger sized areas.

[Table 9]

Second, our main analysis includes students who changed their residence during the survey period. However, since we cannot rule out the possibility that some students relocated to avoid air pollution, as a robustness check, we present the results using a sample of respondents who did not change their residence during the survey period. These results can be found in column (3) of Table 9, and they are largely consistent with our main findings.

Lastly, we sequentially add each pollutant variable other than PM_{10}—such as SO_2, NO_2, CO, and O_3—to our main estimating equation and report the results in Columns (1) through (4) of Table 10, and in column (5), we include all four pollutant variables together. The estimated effects of particulate matter remain robust throughout, with similar magnitudes. Furthermore, the pollutant variables newly included in the model are mostly insignificant across the models, suggesting that the SWB effects we found are primarily attributable to particulate matter.

[Table 10]

5 Conclusion

This paper examines the effect of air pollution (PM_{10}) on adolescents' subjective well-being in South Korea. The main findings suggest a negative effect of air pollution on SWB measured by life satisfaction. Air pollution also has a negative significant effect on positive affect (happiness), and positive significant effect on negative affect (anxiety), although the estimates for negative affect are imprecisely estimated. We also find a non-linear effect, indicating that the negative effect of PM_{10} becomes weaker with a higher level of concentration. The negative impact of PM_{10} on SWB is more concentrated among students with low-income households and male students. As a potential channel for these effects, we examine various daily activities of the

students, such as time spent on sleeping, private tutoring, reading, and digital-entertaining, and find that particulate matter is associated with reduced reading time.

Adolescents are more vulnerable to air pollution exposure compared to adults, and there is a higher likelihood that such exposure will have long-term effects on adult health (Li et al., 2021; Sunyer, 2008). Furthermore, it has been established that even when pollutant concentrations are not high, air pollution can adversely affect the well-being of adolescents (Beavan et al., 2023). Previous research also suggests that non-cognitive skills are an important factor influencing an individual's socioeconomic outcomes (Cunha et al., 2010; Heckman et al., 2006), and our study is one of the rare studies to examine the effect of air pollution on various elements of adolescents' SWB: life satisfaction, positive affect, and negative affect. The findings of this study indicate that air pollution negatively impacts adolescents' subjective well-being. Therefore, our study calls for effective policy measures to reduce adolescents' exposure to air pollution in both schools and homes, as well as to support the well-being of students by providing a healthier environment. In particular, given that our heterogeneity findings indicate that low-SES students are more adversely affected, investments in air filtration in public schools could serve as a progressive policy intervention. These efforts are likely to be especially beneficial for those who may face challenges in mitigating their exposure to pollution.

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Tables and Figures

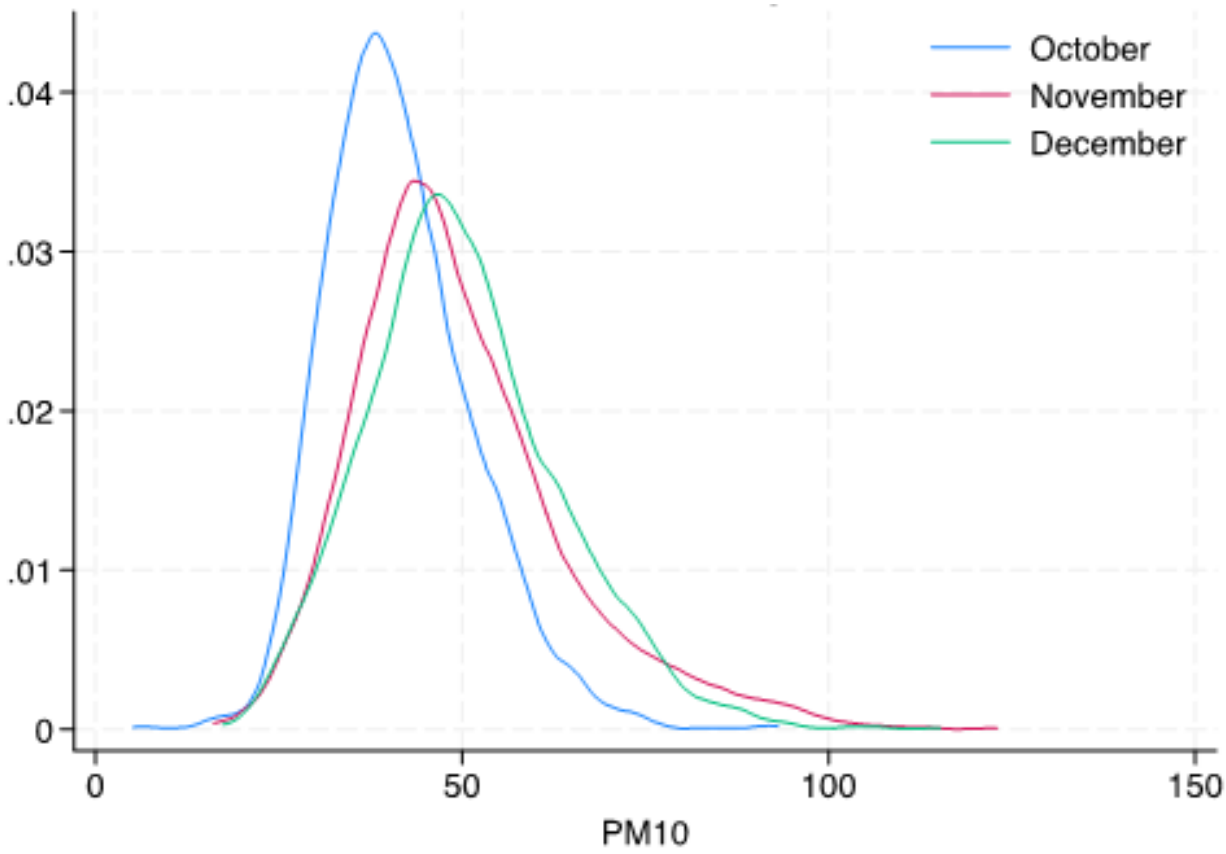


Figure 1: particulate matter Distribution by Month

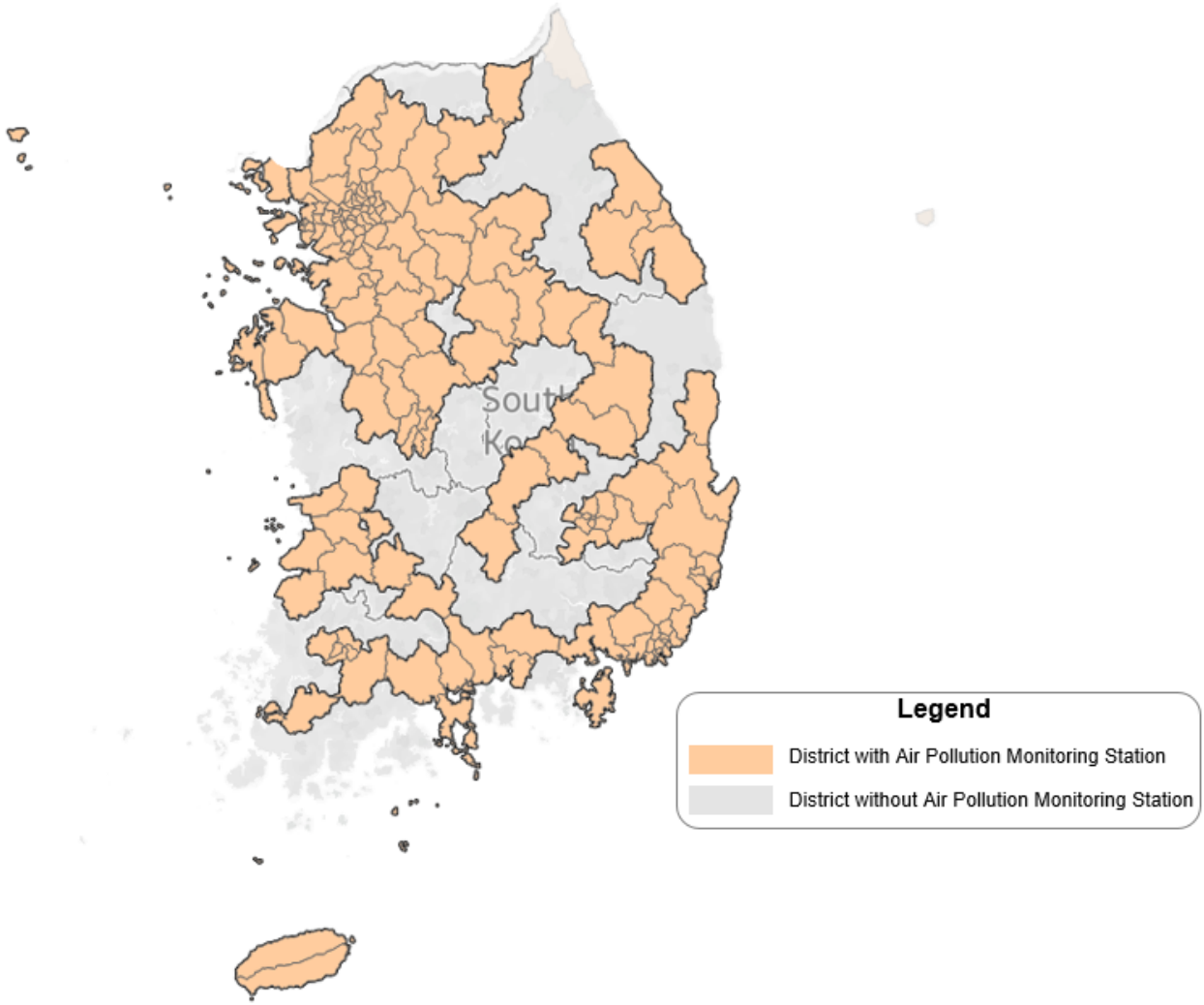


Figure 2: Location of Air Pollution Monitoring Stations

Table 1: Descriptive Statistics of Main Variables, 2010-2016

Variable	Description	Obs.	Mean	Std. Dev.	Min	Max
Panel A. Subjective Well-being Variables						
Life satisfaction (LS)	I think my life is happy	12,374	3.1	0.7	1.0	4.0
Positive affect (happiness)	I enjoy living my life	12,373	3.1	0.7	1.0	4.0
Negative affect (anxiety)	FA of anxiety index	8,116	0.0	1.0	-1.7	3.6
Negative affect (depression)	FA of depression index	8,110	0.0	1.0	-1.5	3.8
Panel B. Time Variables						
Sleeping time (Hour)	Average Sleeping times in a day	12,276	8.2	1.5	2.0	14.2
Private-tutoring time (Hour)	Average private-tutoring times in a day	6,982	1.0	1.3	0.0	10.0
Study-time (Hour)	Average study (homework) times in a day	12,102	2.3	2.0	0.0	17.0
Reading-time (Hour)	Average reading books (not textbooks) times in a day	12,269	0.5	0.7	0.0	7.0
Digital-entertaining time (Hour)	Average Playing on a computer or console and watching TV, DVD times in a day	12,242	3.3	1.9	0.0	16.0
Friend-time (Hour)	The average time of day to play with friends	12,256	1.5	1.4	0.0	10.0
Panel C. Control Variables						
Age	Age	12,374	15.1	1.6	12.0	20.0
Yearly income	Average yearly income in 10,000,000 KRW	12,374	4.6	2.4	0.0	40.0
Parents' years of education	Maximum parent's final years of education	12,374	14.2	3.3	0.0	23.0
Peer relation	Sum of perceived peer relation	12,374	21.7	12.1	0.0	36.0
Supportive parenting	Sum of perceived level of supportive parenting	12,374	11.0	4.6	0.0	16.0
Abuse	Sum of perceived level of abuse	12,374	5.8	3.2	0.0	16.0
Victimization	Sum of perceived level of victimization	12,374	0.1	0.3	0.0	6.0
Panel D. Air Pollution Variables						
PM_{10} (mg/m^3)	Mean of PM_{10} (Oct–Dec)	378	46.6	9.2	22.2	100.9
SO_2 (mg/m^3)	Mean of SO_2 (Oct–Dec)	378	0.005	0.002	0.001	0.013
NO_2 (mg/m^3)	Mean of NO_2 (Oct–Dec)	378	0.026	0.009	0.007	0.050
CO (mg/m3)	Mean of CO (Oct–Dec)	378	0.572	0.155	0.248	1.176
O_3 (mg/m^3)	Mean of O_3 (Oct–Dec)	378	0.018	0.006	0.009	0.050
Panel E. Weather Variables						
Temperature (°C)	Mean of temperature (Oct–Dec)	378	8.4	2.9	0.1	16.8
Rainfall (mm)	Mean of rainfall (Oct–Dec)	378	2.0	3.8	0.0	36.8
Wind speed (m/s)	Mean of wind speed (Oct–Dec)	378	1.9	1.0	0.4	6.1

Note: Summary statistics for 12,374 observations of 3,242 adolescents from seven years of survey are presented.; Statistics in Panels D and E are computed at the district-year level (N = 378), reflecting that air pollution and weather variables vary at the district level and are assigned uniformly to all individuals within the same district-year cell; 10,000 KRW is worth approximately 10 USD; All air pollution and weather variables used the average monthly data of the survey period between October and December, except for the 1st wave of year 2010 where the survey took place between October and November; Multiple questions related to control variables (peer relation, supportive parenting, abuse, victimization) that were summed can be found in Table A1.

Table 2: Effects of particulate matter on SWB

	Model (1)			Model (2)		
	(1)	(2)	(3)	(4)	(5)	(6)
A. Life satisfaction						
$PM_{10}^{\dagger}$	-0.025***	-0.000	-0.001	-0.029	-0.097**	-0.093**
	(0.007)	(0.011)	(0.011)	(0.040)	(0.041)	(0.039)
${PM_{10}^{\dagger}}^2$				0.000	0.009**	0.009**
				(0.004)	(0.004)	(0.003)
Mean	3.076	3.076	3.076	3.076	3.076	3.076
Observations	12,374	12,374	12,374	12,374	12,374	12,374
R-squared	0.001	0.002	0.027	0.001	0.003	0.028
Year FE	N	Y	Y	N	Y	Y
Individual FE	N	Y	Y	N	Y	Y
Controls	N	N	Y	N	N	Y
B. Happiness						
$PM_{10}^{\dagger}$	-0.020***	-0.011	-0.010	0.015	-0.090*	-0.083*
	(0.007)	(0.010)	(0.009)	(0.038)	(0.048)	(0.047)
${PM_{10}^{\dagger}}^2$				-0.003	0.007	0.007
				(0.004)	(0.004)	(0.004)
Mean	3.140	3.140	3.140	3.140	3.140	3.140
Observations	12,373	12,373	12,373	12,373	12,373	12,373
R-squared	0.001	0.003	0.031	0.001	0.003	0.032
Year FE	N	Y	Y	N	Y	Y
Individual FE	N	Y	Y	N	Y	Y
Controls	N	N	Y	N	N	Y

Note: † actual PM_{10} value was divided by 10; Standard errors clustered by location are reported in parentheses; The control variables include individual-level characteristics that may vary over time (such as age and household income) and weather-related variables (temperature, rainfall, and wind speed). See Section 2.2 for further details. In Model (2), the weather-related control variables are included both in levels and squared terms; ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

Table 3: Effects of particulate matter on Positive and Negative Affects

	Positive affect (Life satisfaction)	Positive affect (Happiness)	Negative affect (Anxiety)	Negative affect (Depression)
	(1)	(2)	(3)	(4)
$PM_{10}^{\dagger}$	-0.093**	-0.083*	0.236*	0.082
	(0.039)	(0.047)	(0.137)	(0.099)
${PM_{10}^{\dagger}}^2$	0.009***	0.007	-0.028*	-0.010
	(0.003)	(0.004)	(0.014)	(0.011)
Temperature	0.016*	0.008	-0.028	-0.049**
	(0.010)	(0.011)	(0.020)	(0.019)
Temperature^2	-0.001**	-0.001	0.002**	0.003***
	(0.001)	(0.001)	(0.001)	(0.001)
Rainfall	-0.007**	-0.005	-0.005	0.008*
	(0.003)	(0.003)	(0.006)	(0.004)
Rainfall^2	0.000	0.000*	0.000	-0.000***
	(0.000)	(0.000)	(0.000)	(0.000)
Wind speed	-0.034	0.001	-0.028	-0.059
	(0.034)	(0.027)	(0.066)	(0.045)
Wind speed^2	0.005	0.002	0.005	0.013*
	(0.005)	(0.006)	(0.011)	(0.007)
Age	-0.078***	-0.064***	-0.149***	-0.015
	(0.013)	(0.011)	(0.014)	(0.014)
Yearly income	0.008	0.003	0.003	-0.003
	(0.005)	(0.005)	(0.013)	(0.013)
Parents' years of education	-0.003	0.005	-0.011	0.000
	(0.005)	(0.005)	(0.010)	(0.008)
Peer relation	0.004***	0.005***	-0.017***	-0.036***
	(0.001)	(0.001)	(0.004)	(0.004)
Supportive parenting	0.041***	0.034***	-0.032***	-0.038***
	(0.005)	(0.004)	(0.007)	(0.008)
Abuse	-0.017***	-0.016***	0.057***	0.069***
	(0.004)	(0.004)	(0.007)	(0.007)
Victimization	-0.024	-0.081***	0.052*	0.149***
	(0.026)	(0.022)	(0.026)	(0.032)
Mean	3.076	3.140	0.000	0.000
Observations	12,374	12,373	8,116	8,110
R-squared	0.028	0.032	0.118	0.096
Year FE	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y

Note: † actual PM_{10} value was divided by 10; Standard errors clustered by the district level are reported in parentheses; ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

Table 4: Effects of particulate matter on Time (Potential channels)

	Sleeping time	Private-tutoring time	Study-time	Reading-time	Digital-entertaining time	Friend-time
	(1)	(2)	(3)	(4)	(5)	(6)
$PM_{10}^{\dagger}$	-0.224	0.026	-0.203	-0.105**	-0.024	0.027
	(0.152)	(0.135)	(0.209)	(0.050)	(0.154)	(0.120)
${PM_{10}^{\dagger}}^2$	0.021	-0.002	0.023	0.009*	-0.004	0.000
	(0.013)	(0.010)	(0.017)	(0.004)	(0.012)	(0.010)
Mean	8.226	1.001	2.279	0.515	3.303	1.455
Observations	12,276	6,982	12,102	12,269	12,242	12,256
R-squared	0.346	0.053	0.069	0.026	0.116	0.014
Year FE	Y	Y	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y

Note: † actual PM_{10} value was divided by 10; Standard errors clustered by the district level are reported in parentheses. The control variables include individual-level characteristics that may vary over time (such as age and household income) and weather-related variables (temperature, rainfall, and wind speed). ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

Table 5: Heterogeneity Effects of particulate matter on SWB by Gender

	Male		Female	
	Life satisfaction	Happiness	Life satisfaction	Happiness
$PM^{\dagger}{}_{10}$	-0.127**	-0.107	-0.067	-0.051
	(0.050)	(0.065)	(0.060)	(0.059)
$PM^{\dagger 2}_{10}$	0.013***	0.009	0.004	0.004
	(0.005)	(0.006)	(0.005)	(0.004)
Mean	3.128	3.198	3.016	3.072
Observations	6,680	6,679	5,694	5,694
R-squared	0.028	0.031	0.037	0.038
Year FE	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y
Controls	Y	Y	Y	Y

Note: † actual PM_{10} value was divided by 10. The control variables include individual-level characteristics that may vary over time (such as age and household income) and weather-related variables (temperature, rainfall, and wind speed). Standard errors clustered by the district level are reported in parentheses; ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

Table 6: Effects of particulate matter on Time by gender (Potential channels)

	Male						Female					
	Sleeping time	Private-tutoring time	Study time	Reading time	Digital-entertaining time	Friend time	Sleeping time	Private-tutoring time	Study time	Reading time	Digital-entertaining time	Friend time
$PM^{\dagger}_{10}$	-0.514***	-0.136	-0.133	-0.134*	-0.116	-0.130	0.203	0.305*	-0.383	-0.068	0.130	0.278
	(0.163)	(0.161)	(0.218)	(0.068)	(0.159)	(0.159)	(0.127)	(0.174)	(0.334)	(0.077)	(0.202)	(0.174)
$PM^{\dagger 2}_{10}$	0.049***	0.007	0.015	0.012*	0.004	0.016	-0.017*	-0.020	0.039	0.005	-0.016	-0.022
	(0.014)	(0.013)	(0.017)	(0.006)	(0.014)	(0.014)	(0.010)	(0.012)	(0.026)	(0.006)	(0.015)	(0.014)
Mean	8.249	0.994	2.027	0.510	3.500	1.519	8.200	1.009	2.574	0.522	3.072	1.380
Observations	6,631	3,855	6,527	6,614	6,606	6,614	5,645	3,127	5,575	5,655	5,636	5,642
R-squared	0.355	0.069	0.060	0.023	0.091	0.020	0.342	0.052	0.090	0.036	0.157	0.015
Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

Note: † actual PM_{10} value was divided by 10. The control variables include individual-level characteristics that may vary over time (such as age and household income) and weather-related variables (temperature, rainfall, and wind speed). Standard errors clustered by the district level are reported in parentheses; $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 7: Heterogeneity Effects of particulate matter on SWB by SES

	Low SES		High SES	
	Life satisfaction	Happiness	Life satisfaction	Happiness
$PM^{\dagger}{}_{10}$	-0.075	-0.116*	-0.117	-0.110
	(0.053)	(0.064)	(0.074)	(0.077)
$PM^{\dagger 2}_{10}$	0.008*	0.010*	0.011	0.010
	(0.004)	(0.006)	(0.007)	(0.007)
Mean	3.038	3.109	3.114	3.171
Observations	6,127	6,127	6,247	6,246
R-squared	0.032	0.035	0.025	0.031
Year FE	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y
Controls	Y	Y	Y	Y

Note: We define socioeconomic status (SES) based on the median household income in each survey year. Students are classified as 'High SES' if their household income is above the year-specific median and 'Low SES' otherwise. † actual PM_{10} value was divided by 10. The control variables include individual-level characteristics that may vary over time (such as age and household income) and weather-related variables (temperature, rainfall, and wind speed). Standard errors clustered by the district level are reported in parentheses; ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

Table 8: Effects of particulate matter on Time by SES (Potential channels)

	Low SES						High SES					
	Sleeping time	Private-tutoring time	Study time	Reading time	Digital-entertaining time	Friend time	Sleeping time	Private-tutoring time	Study time	Reading time	Digital-entertaining time	Friend time
$PM^{\dagger}{}_{10}$	-0.163	-0.225*	-0.512*	-0.137***	0.001	0.130	-0.331*	0.307*	0.121	-0.010	-0.011	0.024
	(0.146)	(0.121)	(0.265)	(0.049)	(0.230)	(0.156)	(0.184)	(0.168)	(0.250)	(0.079)	(0.175)	(0.146)
$PM^{\dagger 2}_{10}$	0.017	0.019**	0.047**	0.012***	-0.006	-0.005	0.031*	-0.022	-0.005	-0.001	-0.004	-0.007
	(0.011)	(0.009)	(0.020)	(0.004)	(0.019)	(0.012)	(0.017)	(0.014)	(0.021)	(0.007)	(0.016)	(0.012)
Mean	8.355	0.734	1.904	0.488	3.588	1.590	8.101	1.261	2.652	0.542	3.024	1.324
Observations	6,070	3,443	6,029	6,077	6,051	6,056	6,206	3,539	6,073	6,192	6,191	6,200
R-squared	0.286	0.053	0.065	0.025	0.098	0.018	0.378	0.050	0.074	0.032	0.137	0.021
Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

Note: † actual PM_{10} value was divided by 10. The control variables include individual-level characteristics that may vary over time (such as age and household income) and weather-related variables (temperature, rainfall, and wind speed). Standard errors clustered by the district level are reported in parentheses; $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 9: Robustness Checks: Effects of particulate matter on SWB by Area size and Non-movers sample

	Small areas		Large areas		Non-movers	
	Life satisfaction	Happiness	Life satisfaction	Happiness	Life satisfaction	Happiness
${PM^{\dagger}}_{10}$	-0.176*	-0.342***	-0.076*	-0.071	-0.087**	-0.083*
	(0.103)	(0.085)	(0.043)	(0.053)	(0.041)	(0.049)
${PM^{\dagger 2}}_{10}$	0.017	0.036***	0.007**	0.005	0.008**	0.007
	(0.011)	(0.009)	(0.003)	(0.004)	(0.003)	(0.004)
Mean	3.113	3.163	3.055	3.127	3.078	3.141
Observations	4,529	4,529	7,845	7,844	11,828	11,827
R-squared	0.037	0.041	0.027	0.031	0.029	0.034
Year FE	Y	Y	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y

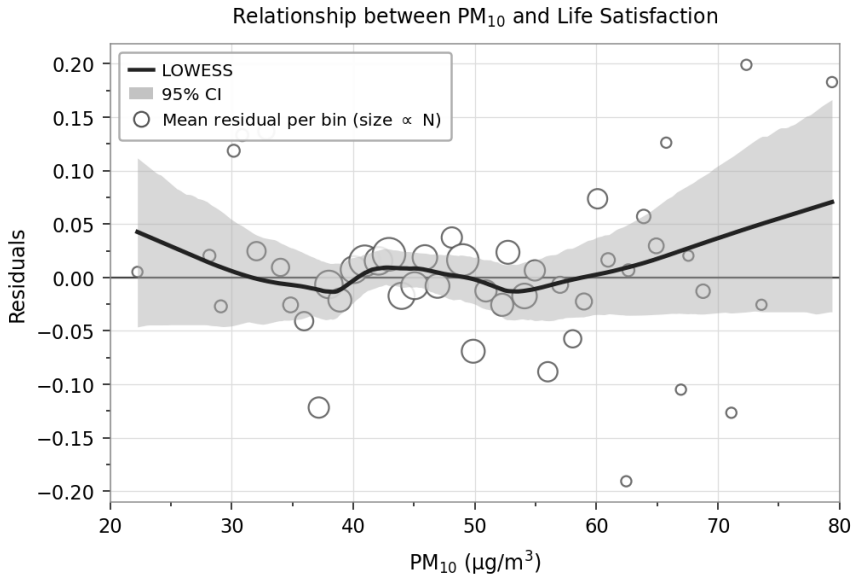
Note: Small areas include 1 special city (Seoul), 6 metropolitan cities (Busan, Daegu, Incheon, Gwangju, Daejeon, Ulsan), and 1 special self-governing city (Sejong).; Non-movers represents the results from analyzing only students who remained in the same residence throughout the analysis period.; The control variables include individual-level characteristics that may vary over time (such as age and household income) and weather-related variables (temperature, rainfall, and wind speed).; † actual PM_{10} value was divided by 10; Standard errors clustered by the district level are reported in parentheses; ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

Table 10: Robustness Checks: Effects of particulate matter on SWB including all Air Pollutants

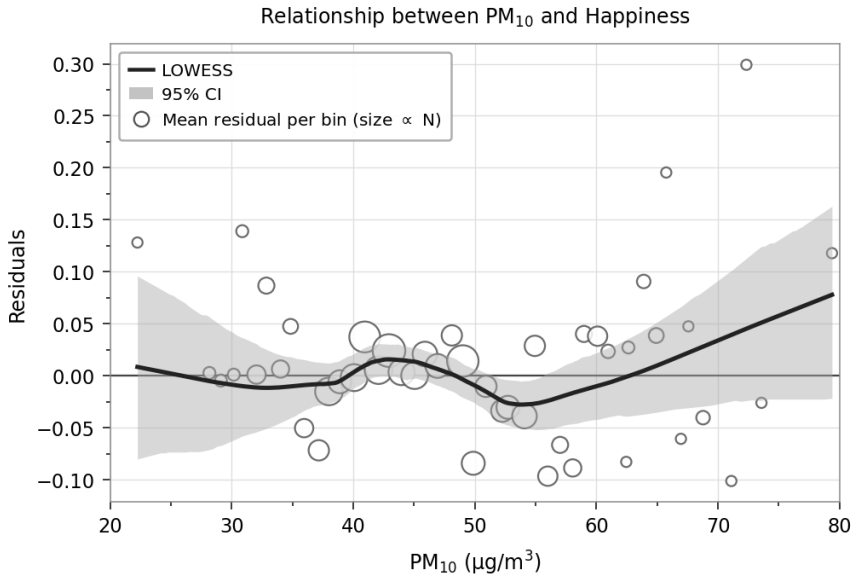
	(1)	(2)	(3)	(4)	(5)
Panel A: Life Satisfaction					
$PM^{\dagger}{}_{10}$	-0.095**	-0.091**	-0.107***	-0.092**	-0.109***
	(0.039)	(0.038)	(0.038)	(0.039)	(0.036)
$PM^{\dagger 2}_{10}$	0.009***	0.008**	0.010***	0.009**	0.010***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
SO_2	-29.840				-25.318
	(33.531)				(10.105)
SO_2^2	2876.927				2286.680
	(2985.233)				(2857.840)
NO_2		-1.574			0.037
		(9.203)			(9.056)
NO_2^2		7.593			-16.880
		(143.254)			(140.002)
CO			-0.315		-0.266
			(0.249)		(0.238)
CO^2			0.349**		0.319**
			(0.157)		(0.154)
O_3				8.149	8.229
				(9.019)	(8.922)
O_3^2				-97.319	-81.353
				(128.295)	(128.503)
Observations	12,374	12,374	12,374	12,374	12,374
R-squared	0.028	0.028	0.029	0.028	0.029
Panel B: Happiness					
$PM^{\dagger}{}_{10}$	-0.087*	-0.076	-0.089*	-0.083*	-0.083*
	(0.045)	(0.049)	(0.047)	(0.049)	(0.048)
$PM^{\dagger 2}_{10}$	0.007*	0.006	0.007*	0.007	0.007
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
SO_2	-49.361*				-54.475**
	(26.723)				(25.035)
SO_2^2	4964.575**				5350.410***
	(2171.479)				(2007.579)
NO_2		-8.862			-11.427
		(7.973)			(8.541)
NO_2^2		123.671			158.055
		(141.718)			(149.729)
CO			-0.019		0.213
			(0.290)		(0.318)
CO^2			0.075		-0.092
			(0.188)		(0.196)
O_3				3.777	3.490
				(8.716)	(8.370)
O_3^2				-42.161	-33.002
				(153.218)	(143.107)
Observations	12,373	12,373	12,373	12,373	12,373
R-squared	0.033	0.032	0.032	0.032	0.033
Year FE	Y	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y

Note: † actual PM_{10} value was divided by 10. The control variables include individual-level characteristics that may vary over time (such as age and household income) and weather-related variables (temperature, rainfall, and wind speed). Standard errors clustered by the district level are reported in parentheses; $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Appendix



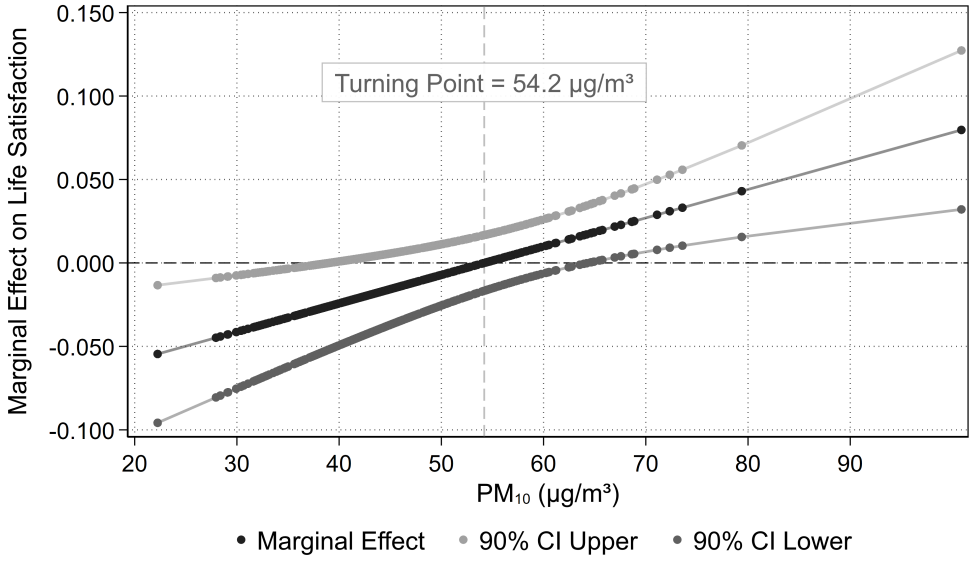
(a) Life Satisfaction



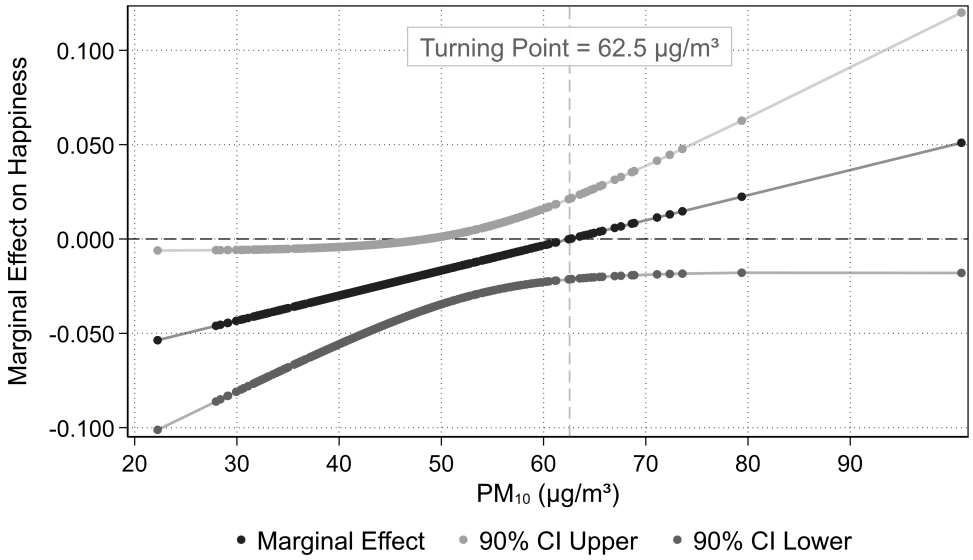
(b) Happiness

Note: Each circle represents the mean residual within an integer PM_{10} bin, with circle size proportional to the number of observations in the bin. The solid curve is a LOWESS smoother estimated with a bandwidth of 0.5, where each observation is weighted by bin size. The shaded region denotes pointwise 95% confidence intervals constructed from 400 bootstrap replications. Residuals are obtained from a regression of life satisfaction or happiness on PM_{10} with district-level and year fixed effects.

Figure A1: Nonlinear Relationship between PM_{10} and Subjective Well-being



(a) Life Satisfaction



(b) Happiness

Note: Marginal effects of PM_{10} on subjective well-being with 90% confidence intervals. The circles indicate turning points where marginal effects equal zero. Standard errors are clustered at the district level.

Figure A2: Marginal Effects of PM_{10} on Subjective Well-being

Table A1: Questionnaire

Category	Questions
Anxiety	I sometimes nitpick on small things I sometimes bother what others do If someone does not allow me to do what I want, I argue with them I argue with small matters I can get angry all day long I sometimes cry for no reason
Depression	I have little energy I feel unhappy or sad and depressed I have a lot of worries I feel like I want to die I cry a lot When something goes wrong, I often think it is because of me I feel lonely I am not interested in anything I feel like the future is not hopeful Everything seems hard to do
Peer relation	My friends respect my idea when they talk to me My friends listen to what I say I talk to my friends about my woes and problems My friends understand me very well I can tell my friends when I want to tell them what I feel inside I trust my friends I want to make other friends instead of my current friends. Even when I am with my friends, I feel lonely and alone My friends don't know how I'm doing these days
Supportive parenting	They consider me more important than work They are genuinely interested in my school life They always keep my body, clothes and blankets clean If I'm very sick, they give me proper treatment.
Abuse	They scold me more than what I do wrong They try to hit me unconditionally if I do something wrong They often hit me severely to the point I get bruises or scars on my body They often assault with insults
Victimization	Severely teased or ridiculed Bullied Severely assaulted Blackmailed Stolen of money or possessions Was sexually assaulted or harassed Severely insulted

Note: All questions ranging from Anxiety to Abuse are scaled on a 4-point Likert scale where 1 indicates strongly agree, 2 indicates somewhat agree, 3 indicates somewhat disagree, and 4 indicates strongly disagree; all the survey questions were reversely coded; Answers to questions in the victimization category are represented by a dummy variable, taking the value of 1 if the respondent has experienced the event in question, and 0 otherwise.

Table A2: Factor Analysis

Factor		Factor Loadings
Anxiety	I sometimes nitpick on small things	0.7466
	I sometimes bother what others do	0.6928
	If someone does not allow me to do what I want I argue with them	0.7321
	I argue with small matters	0.7831
	I can get angry all day long	0.7494
	I sometimes cry for no reason	0.6074
Depression	I have little energy	0.6954
	I feel unhappy or sad and depressed	0.7819
	I have a lot of worries	0.6893
	I feel like I want to die	0.6921
	I cry a lot	0.6334
	When something goes wrong I often think it is because of me	0.7092
	I feel lonely	0.7310
	I am not interested in anything	0.7360
	I feel like the future is not hopeful	0.7225
	Everything seems hard to do	0.8108

Note: The answers are rated on a Likert scale ranging from 1 = "strongly agree" to 4 = "strongly disagree."; all the survey questions were reversed coded; We extract one optimal factor by factor analysis. This factor contains Eigenvalues > 1 and explains $> 50\%$ of the cumulative variance and Kaiser–Meyer–Olkin (KMO) > 0.6.

Table A3: WHO air quality guidelines and interim targets of PM_{10}

PM_{10} ($\mu g/m^3$)	Annual	24-hour	Basis for selected level for annual mean
Interim target-1 (IT-1)	70	150	These levels are associated with about a 15% higher long-term mortality risk relative to the AQG level.
Interim target-2 (IT-2)	50	100	In addition to other health benefits, these levels lower the risk of premature mortality by approximately 6% [2–11%] relative to the IT-1 level.
Interim target-3 (IT-3)	30	75	In addition to other health benefits, these levels reduce the mortality risk by approximately 6% [2–11%] relative to the IT-2 level.
Air quality guideline (AQG)	20	50	These are the lowest levels at which total cardiopulmonary and lung cancer mortality have been shown to increase with more than 95% confidence in response to long-term exposure to $PM_{2.5}$.

Table A4: Balance Test

	Panel A						Panel B					
	(1) Yearly income	(2) Parents' years of education	(3) Peer relation	(4) Supportive parenting	(5) Abuse	(6) Victimization	(1) Yearly income	(2) Parents' years of education	(3) Peer relation	(4) Supportive parenting	(5) Abuse	(6) Victimization
$PM_{10}^{\dagger}$	0.730***	1.120***	-0.437	-0.023	0.161	0.009	0.115	0.126	-0.365	-0.004	0.071	0.037
	(0.276)	(0.372)	(0.667)	(0.134)	(0.149)	(0.017)	(0.139)	(0.099)	(1.486)	(0.108)	(0.138)	(0.023)
${PM_{10}^{\dagger}}^2$	-0.066***	-0.086***	0.043	0.002	-0.010	-0.000	-0.017	-0.010	0.042	-0.002	-0.010	-0.003
	(0.022)	(0.029)	(0.062)	(0.010)	(0.012)	(0.001)	(0.014)	(0.010)	(0.101)	(0.009)	(0.010)	(0.002)
Observations	12,374	12,374	12,374	12,374	12,374	12,374	12,374	12,374	12,374	12,374	12,374	12,374
R-squared	0.008	0.007	0.325	0.825	0.481	0.020	0.023	0.001	0.339	0.887	0.624	0.028
Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Individual FE	N	N	N	N	N	N	Y	Y	Y	Y	Y	Y
Weather controls	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

Note: † actual PM_{10} value was divided by 10; Standard errors clustered by the district level are reported in parentheses; ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

Table A5: Heterogeneity Effects of particulate matter on Negative Affects

	(1)		(2)	
	Anxiey	Depression	Anxiey	Depression
Panel A: By Gender				
	Male		*Female*	
$PM^{\dagger}{}_{10}$	0.100	0.033	0.412	0.050
	(0.133)	(0.155)	(0.271)	(0.204)
$PM^{\dagger 2}_{10}$	-0.014	-0.008	-0.044	-0.001
	(0.015)	(0.016)	(0.029)	(0.022)
Mean	-0.038	-0.126	0.045	0.147
Observations	4,367	4,364	3,749	3,746
R-squared	0.123	0.085	0.121	0.124
Panel B: By Socioeconomic Status (SES)				
	Low SES		*High SES*	
$PM^{\dagger}{}_{10}$	0.128	0.010	0.242	0.121
	(0.220)	(0.166)	(0.213)	(0.145)
$PM^{\dagger 2}_{10}$	-0.018	-0.003	-0.027	-0.015
	(0.023)	(0.018)	(0.022)	(0.015)
Mean	0.035	0.057	-0.035	-0.056
Observations	4,035	4,031	4,081	4,079
R-squared	0.032	0.035	0.025	0.031
Year FE	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y
Controls	Y	Y	Y	Y

Note: † Actual PM_{10} value was divided by 10. Panel B defines socioeconomic status (SES) based on the median household income in each survey year; students are classified as 'High SES' if their household income is above the year-specific median and 'Low SES' otherwise. The control variables include individual-level characteristics that may vary over time (such as age and household income) and weather-related variables (temperature, rainfall, and wind speed). Standard errors clustered by the district level are reported in parentheses; ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

Table A6: Balance Test: Residential Mobility

	Pr(Moving)	
	(1)	(2)
$PM_{10}^{\dagger}$	0.017	0.057
	(0.022)	(0.038)
$PM_{10}^{\dagger 2}$	-0.002	-0.005
	(0.002)	(0.004)
Observations	10,802	10,802
R-squared	0.000	0.028
Year FE	N	Y
Individual FE	N	Y
Controls	N	Y

Note: The dependent variable is a dummy equal to one if the student changed their district of residence in a given year.; The base year 2010 is excluded since the mobility indicator is mechanically zero by construction. † Actual PM_{10} value was divided by 10. Standard errors clustered at the district level are reported in parentheses; ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.